Notes

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1 Failure on high-dimensional circular data

We have shown that flow matching with path integration works well for high-dimensional non-circular data; here we examine whether it works for *circular* tuning on θ using the toy family below.

1.1 Toy dataset

We use a synthetic conditional model for a scalar stimulus θ and a multivariate observation $x \in \mathbb{R}^d$ with $d \geq 2$. Stimuli are drawn independently from a uniform density on a bounded interval,

$$\theta \sim \mathrm{Uniform}(\theta_{\mathrm{low}}, \theta_{\mathrm{high}}). \tag{1}$$

Conditional mean (tuning curve). For each coordinate $j \in \{1, \ldots, d\}$ fix a phase $\phi_j = 2\pi(j-1)/d$. Given θ, the mean of x_j is a cosine in θ,

$$\mu_j(\theta) = A\cos(\omega\theta + \phi_j), \qquad j = 1, \ldots, d, \tag{2}$$

with amplitude $A > 0$ and frequency $\omega > 0$ on the stimulus axis (in the default bundle, $A = \omega = 1$). The conditional mean vector is $\mu(\theta) = (\mu_1(\theta), \ldots, \mu_d(\theta))^\mathsf{T}$.

Observation noise. Conditionally on θ, the observation is multivariate Gaussian with diagonal covariance. Let $\sigma_{\mathrm{base},j} > 0$ be nonnegative baseline scales across neurons; in the usual configuration they interpolate linearly in j between fixed σ_1 and σ_2, and often $\sigma_1 = \sigma_2 = \sigma$ so all coordinates share the same σ. Let α be a fixed activity–variance coupling strength (the arithmetic mean of two nonnegative hyperparameters in the implementation), and let $\varepsilon > 0$ be a negligible variance floor. The per-neuron conditional variances are

$$v_j(\theta) = d\,\sigma_{\mathrm{base},j}^2\left(1 + \alpha\,|\mu_j(\theta)|\right) + \varepsilon, \qquad j = 1, \ldots, d, \tag{3}$$

with independence across neurons,

$$x \mid \theta \sim \mathcal{N}\big(\mu(\theta),\, \mathrm{diag}(v_1(\theta), \ldots, v_d(\theta))\big). \tag{4}$$

Thus noise magnitude increases with $|\mu_j(\theta)|$, and each neuron's noise standard deviation scales proportionally to $\sqrt{d}$ for fixed $\sigma_{\mathrm{base},j}$, α, and $\mu_j(\theta)$. The leading factor d matches the "$\sqrt{d}$ noise scaling" construction used elsewhere in these notes: as d grows, per-coordinate variance grows with d so that dimension alone does not drive the observation toward an effectively noiseless, trivially informative regime for θ.

1.2 Flow matching with path integration failed on 3D circular dataset

Conditional flow matching trains a stimulus-conditional velocity by regressing on the affine bridge law in (**??**)–(**??**). The Fisher-score surrogate uses the Gaussian probability path (**??**) and the same velocity-to-score map (**??**) as in that note. Figure 1 shows the outcome on the $d = 3$ circular toy above. The Hellinger matrix (top row) stays far from the approximate ground truth: off-diagonal Spearman correlation with that reference barely increases with training size n. The decoding matrix (bottom row) does better but still does not match the reference at the largest n shown.

This behavior is expected in hindsight: turning a learned velocity into a score and integrating along θ is an indirect route when the stimulus acts through periodic tuning, so approximation error can accumulate.

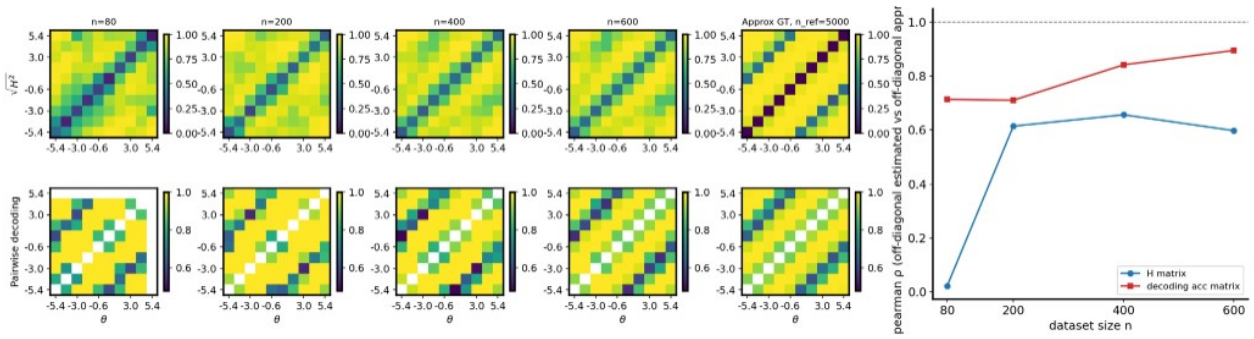

Figure 1: Flow matching with path integration on the $d = 3$ circular (cosine Gaussian $\sqrt{d}$) toy. Top row: estimated $\sqrt{H^2}$ RDMs; bottom row: pairwise decoding matrices; columns are training sizes n and an approximate ground truth from a large reference draw. Right: Spearman correlation between off-diagonal entries of each estimated matrix and the corresponding approximate ground truth, as a function of n (Hellinger in blue, decoding in red).

1.3 Failure on high-dimensional circular data using exact likelihood computation with flow matching on θ space and x space

Flow matching also supports a more direct route: integrate the divergence of the learned velocity along an ODE to obtain a log-density, without converting to a score or integrating a score along θ. Let v_ϕ denote the learned velocity field on observation space. Then the density carried along the flow obeys

$$\frac{\mathrm{d}}{\mathrm{d}t} \log p_t(x_t \mid \theta) = -\nabla_{x_t} \cdot v_\phi(x_t, t \mid \theta) \quad \text{for } x_t \in \mathbb{R}^d, \tag{5}$$

and integrating in time gives the endpoint log-likelihood

$$\log p(x_1 \mid \theta) = \log p_0(x_0) - \int_0^1 \big(\nabla_{x_t} \cdot v_\phi\big)(x_t, t \mid \theta)\, \mathrm{d}t, \tag{6}$$

along the trajectory of $\frac{\mathrm{d}}{\mathrm{d}t} x_t = v_\phi(x_t, t \mid \theta)$, with $\log p_0$ a standard Gaussian on $\mathbb{R}^d$ at $t = 0$. One may evaluate $\log p(x \mid \theta)$ either by the x-space construction in the next paragraph ("x-likelihood"), or by the θ-field construction after it ("θ-likelihood").

x-likelihood. Here the setup is the most direct: fit *one* conditional flow-matching model on observation space with $\frac{\mathrm{d}}{\mathrm{d}t} x_t = v_\phi(x_t, t \mid \theta)$, where θ is fixed when training and at evaluation time. For each stimulus value θ_j and each observation x_i, run the ODE likelihood in (5)–(6) with that θ_j to obtain $\log p(x_i \mid \theta_j)$. There is no separate prior flow on θ and no Bayes step: the matrix with entries $C_{ij} = \log p(x_i \mid \theta_j)$ already targets the conditional log-likelihoods needed downstream (with the usual ΔL diagonal centering applied in the same way as for other C matrices in this project).

θ-likelihood. The idea is to run the same likelihood formula on the *stimulus* θ instead of on x, using two separate flow fits. The first model ignores the observation and follows $\frac{\mathrm{d}}{\mathrm{d}t} \theta_t = v_{\text{prior}}(\theta_t, t)$; it supplies the prior density $\log p(\theta)$. The second model conditions on a fixed x and follows $\frac{\mathrm{d}}{\mathrm{d}t} \theta_t = v_{\text{post}}(\theta_t, t \mid x)$; it supplies $\log p(\theta \mid x)$. In both cases the endpoint log-density is

$$\log p(\theta_1) = \log p_0(\theta_0) - \int_0^1 \frac{\partial v}{\partial \theta}(\theta_t, t)\, \mathrm{d}t, \tag{7}$$

where θ_t solves the ODE for the active field, v is either v_{prior} or $v_{\text{post}}(\cdot \mid x)$, and $\log p_0(\theta_0) = -\frac{1}{2}\big(\theta_0^2 + \log(2\pi)\big)$ for a standard normal base on $\mathbb{R}$. Subtracting the prior evaluation from the posterior evaluation at the same θ yields $\log p(\theta \mid x) - \log p(\theta)$, which Bayes' rule rewrites as $\log p(x \mid \theta) - \log p(x)$. For a fixed observation x_i, that missing term is the same for every candidate θ_j in a row of the pairwise matrix $C_{ij} = \log p(\theta_j \mid x_i) - \log p(\theta_j)$. The subsequent ΔL step removes such row-wise constants, so the contrasts line up with differences in $\log p(x_i \mid \theta_j)$.

Fourier features for θ in the velocity networks. Besides choosing whether likelihoods are accumulated in x-space or in θ-space, one can change how θ is presented to the velocity network. The default is to concatenate the raw scalar θ (together with time and, depending on the field, x or other inputs) as inputs to a standard multilayer architecture. An alternative is a *Fourier-feature* parameterization: map θ to a vector that may include a constant term, the identity term θ, and, for harmonics $k = 1, \ldots, K$ and a fixed angular scale $\omega > 0$, the pairs $\sin(k\omega\theta)$ and $\cos(k\omega\theta)$, then pass that vector through the same style of network. This makes periodic structure in θ easier to represent and is available for both posterior and prior θ-flows, and analogously for the conditional x-flow where θ enters only as a conditioning variable; the harmonic count K and ω are part of the experimental design alongside architecture width and depth.

1.4 Failure of methods on high-dimensional circular data: results

Figure 2. I tried also using Spearman correlation, does work.

In fact, even increasing the number of data the flow matching method still does not work well, I need to check it.

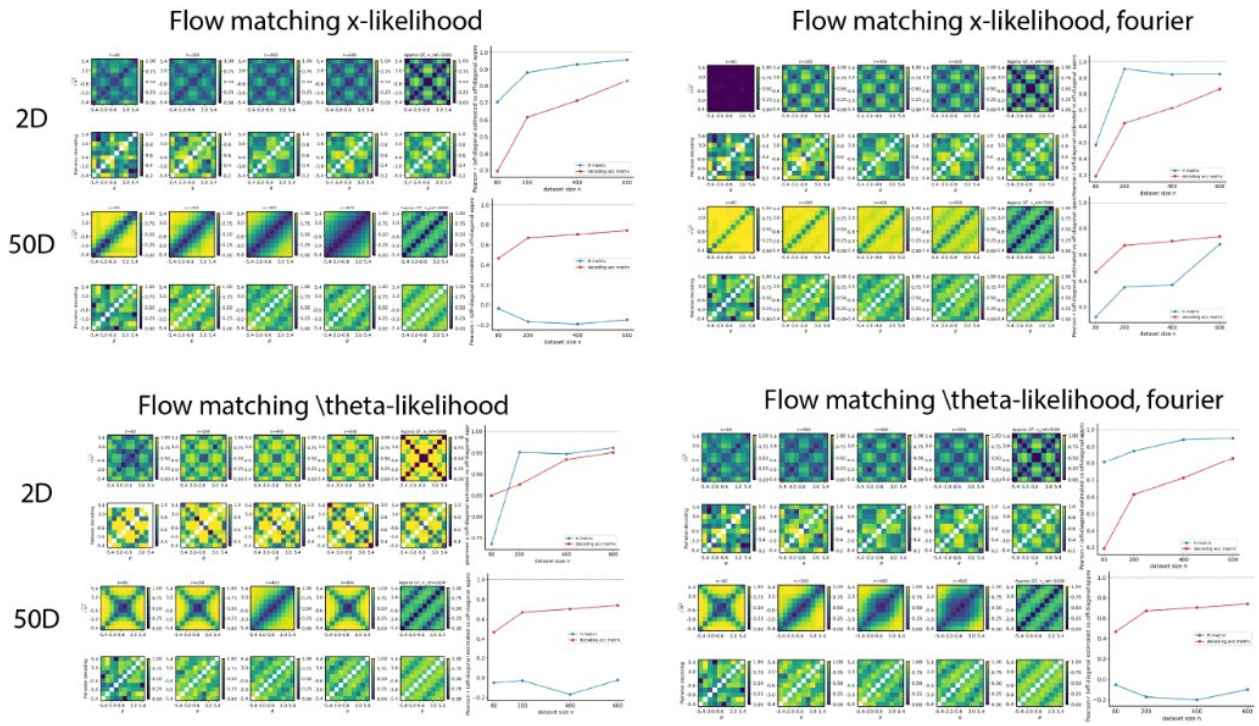

Figure 2: Flow-matching likelihoods: x-likelihood (top left), x-likelihood with Fourier features on θ (top right), θ-likelihood (bottom left), θ-likelihood with Fourier features (bottom right). Within each panel, $d = 2$ then $d = 50$; for each d, top row $\sqrt{\Sigma^{-1}}$–style matrix, bottom row Pearson decoding matrices, columns $n \in \{80, 200, 400, 600\}$ and an approximate ground truth ($n_{\text{ref}} = 5000$). Side plots: Pearson r (diagonal of estimate versus off-diagonal match to the reference) versus n for the H/precision matrix (blue) and the decoding-accuracy matrix (red).